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Explaining increasing fluidization due to activity in a noisy-dense-active system using renormalization group.

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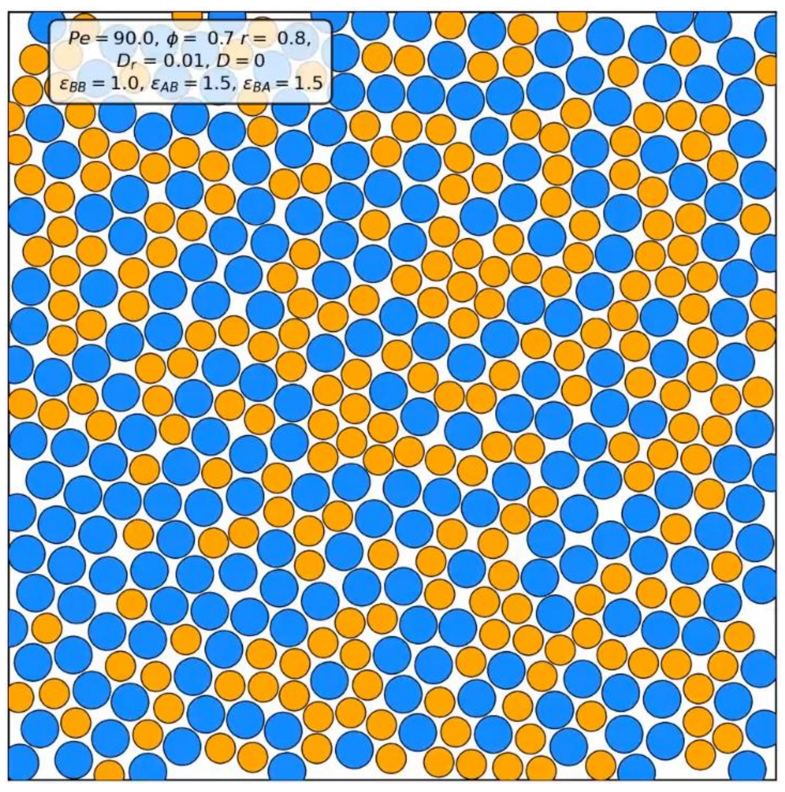
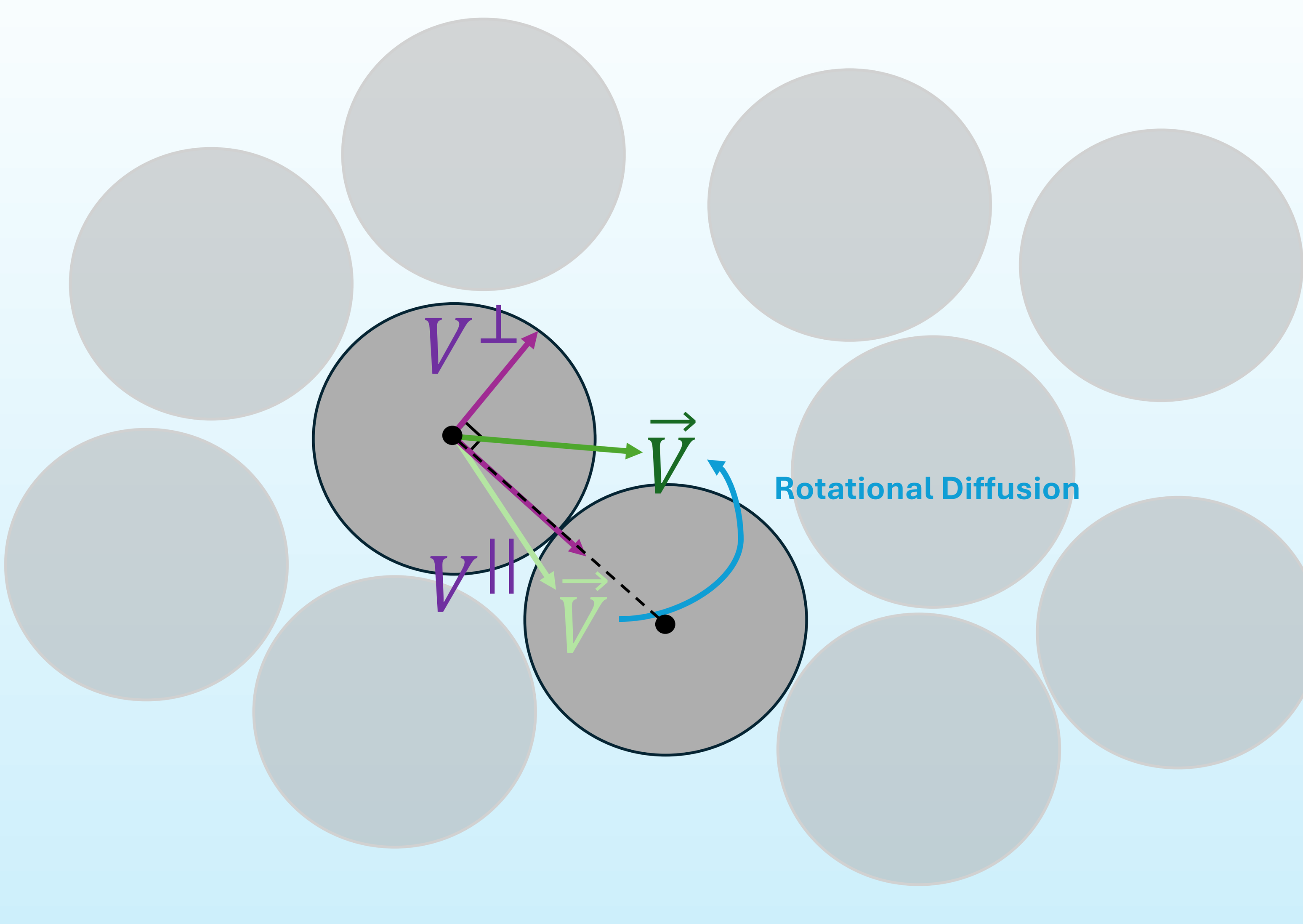
The study of active glasses, meaning, **dense active systems where each agent has self-propulsion** is fundamental to understand the behavior of biological systems, where it has been observed both in the intracellular and intercellular dynamics a behavior corresponding to a glassy system. Likewise, the way in which glassy systems relax stresses is very similar to T1 transitions that appear for example in living cell tissues. Besides biology, there exist synthetic glassy active systems based on driven colloids or granular media.

A simulation of a dense active system can be view by analyzing the QR-code. Next to it there is a screenshot of the video.



Many studies based on agent simulations have reported an increment of the mean squared displacement of tracers as a function of activity **[1,2,3]**. Additionally, it has been shown that glassy phase transition occurs at a packing fraction close to the *random close packing* **[4]**, which represents the disordered state with the highest possible density. Such results have been interpreted as an **effective fluidization of the dense active system**. Currently, a theoretical development that explain this phenomenon is missing in the literature. This is the motivation to pose the following questions:

1. **What does renormalization group flow tell us?**
2. **Could fluidization of dense active systems be explained by longitudinal and transverse fluxes?**

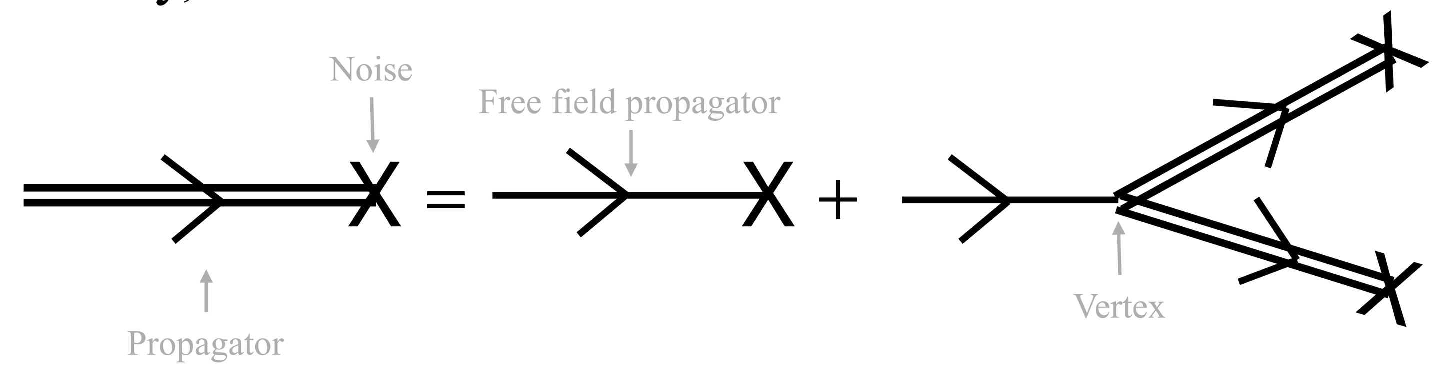


Starting from a Langevin equation for each agent in the system, one can find a field equation for the density following the Dean method **[5]**. To simplify the analysis, we can expand in gradients some terms which yield, after re-defining some parameters, the **Active Model B+ (AMB+) [6]**, namely, a **continuity equation** with a current given by

$$\vec{J} = -\nabla\left[a\rho + \lambda(\nabla\rho)^2\right] + \zeta\nabla\rho\nabla^2\rho + \sqrt{2D\rho}\,\vec{\xi}$$

where the λ and ζ terms generate are non-variational currents. In particular, λ produces just longitudinal currents, while ζ both longitudinal and **transverse**.
The equation for the density perturbation in Fourier space, diagrammatically, takes the form

$\delta\tilde{\rho}(k) :=$



By using the methodology of **Renormalization Group**, computing exact propagators, vertices and noise at **1-loop** one can find the following flux equations for $d' = 2$ and $\lambda = 0$:

Dynamical exponent

$$\frac{d\ln(a)}{d\ell} = z - 2 = 0$$

Renormalization exponent

$$\frac{d\ln(D)}{d\ell} = 3y_\rho - 3z$$

RG cut-off

$$\frac{d\ln(\zeta)}{d\ell} = z + y_\rho - 4 - \zeta^2\left[\frac{D\rho_0}{2\pi\, a^3}\Lambda^5\right] = 0$$

$$\frac{1}{D}\frac{dD}{d\ell} = \frac{3\,\Lambda^5}{32\pi}\frac{D\rho_0}{a^3}\zeta^2 > 0$$

D is relevant and renormalizes due to ζ!

Additionally, and since we are interested in longitudinal and transverse fluxes, one could study the structure factors to understand how current fluctuations are distributed in Fourier space. At **1-loop** and in the **hydrodynamical limit** ($k \to 0$), with $d' = 2$ and λ arbitrary one gets:

$$S_{\delta J^{||}} := \langle \delta J^{||}\delta J^{||}\rangle = S_0 + k^2\frac{(D\rho_0)^2}{a^3}\frac{(8\lambda^2+\zeta^2)}{64\pi}\Lambda^4 + O(k^4)$$

$$S_{\delta J^{\perp}} := \langle \delta\overrightarrow{J^{\perp}}\cdot\delta\overrightarrow{J^{\perp}}\rangle = S_0 + k^2\frac{(D\rho_0)^2}{a^3}\frac{\zeta^2}{4\times 64\pi}\Lambda^4 + O(k^4)$$

Notice that the 1-loop correction depends on the non-variational parameters of the **AMB+**, with λ contributing just to the longitudinal part, while ζ contributing to both as expected.

Key takeaways and future work:

- Noise magnitude is a relevant variable and monotonically increasing function of the active parameter ζ. Increasing activity, increases noise in the hydrodynamical limit which could facilitate the motion of particles in a dense system.
- The ζ interaction is the one responsible perpendicular currents as shown in the computation of the structure factors, so it should be the one in charge of the fluidization of the system.
- It would be interesting to compute the effective diffusion coefficient and compare with numerical simulations and experiments.

References:
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